

Hospital Data Analysis and Predictive Modelling

INTRODUCTION

The purpose of this healthcare dataset from Kaggle ([Link](#)), is to provide data science, machine learning, and data analysis enthusiasts with a useful resource. Because it is made to resemble actual healthcare data, users can practice, hone, and demonstrate their data manipulation and analytic abilities within the framework of the healthcare sector.

Dataset Information:

Each column provides specific information about the patient, their admission, and the healthcare services provided, making this dataset suitable for various data analysis and modelling tasks in the healthcare domain. Here's a brief explanation of each column in the dataset -

- **Name:** This column represents the name of the patient associated with the healthcare record.
- **Age:** The age of the patient at the time of admission, expressed in years.
- **Gender:** Indicates the gender of the patient, either "Male" or "Female."
- **Blood Type:** The patient's blood type, which can be one of the common blood types (e.g., "A+", "O-", etc.).
- **Medical Condition:** This column specifies the primary medical condition or diagnosis associated with the patient, such as "Diabetes," "Hypertension," "Asthma," and more.
- **Date of Admission:** The date on which the patient was admitted to the healthcare facility.
- **Doctor:** The name of the doctor responsible for the patient's care during their admission.
- **Hospital:** Identifies the healthcare facility or hospital where the patient was admitted.
- **Insurance Provider:** This column indicates the patient's insurance provider, which can be one of several options, including "Aetna," "Blue Cross," "Cigna," "UnitedHealthcare," and "Medicare."
- **Billing Amount:** The amount of money billed for the patient's healthcare services during their admission. This is expressed as a floating-point number.
- **Room Number:** The room number where the patient was accommodated during their admission.
- **Admission Type:** Specifies the type of admission, which can be "Emergency," "Elective," or "Urgent," reflecting the circumstances of the admission.
- **Discharge Date:** The date on which the patient was discharged from the healthcare facility, based on the admission date and a random number of days within a realistic range.

- **Medication:** Identifies a medication prescribed or administered to the patient during their admission. Examples include "Aspirin," "Ibuprofen," "Penicillin," "Paracetamol," and "Lipitor."
- **Test Results:** Describes the results of a medical test conducted during the patient's admission. Possible values include "Normal," "Abnormal," or "Inconclusive," indicating the outcome of the test.

Methodology

1. Data Understanding

- After getting the required data from the sources, we understand what business/client wants from the particular data.
- In our case, we have to determine which hospitals have high admission rates, and the common medications and diagnosis. Thus, after understanding, we separate the unnecessary data from the out main data.

2. Data Preparation

- In this stage, we sort the data. Mainly we clean the data for better analysis, we removed unnecessary column for example to get a good output for each disease. Like that, we prepare data using data analysis tools and methods.

3. Data Cleaning

- During Data Preparation, we clean the data and sort out the table for only hospitals, with their doctors list. We clean the data from unwanted things, leaving us the rest for further data analysis.

4. Data Analysis

```
[1]: import pandas as pd
import numpy as np
```

```
[3]: df = pd.read_csv('healthcare_dataset.csv')
```

```
[7]: df
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication
0	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamc
1	Leslie TErRy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327287	265	Emergency	2019-08-26	Ibuprofe
2	DaNnY sMitH	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096079	205	Emergency	2022-10-07	Aspiri
3	andrEw waTtS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.782410	450	Elective	2020-12-18	Ibuprofe
4	adriENNE bEll	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317814	458	Urgent	2022-10-09	Penicilli
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
55495	eLIZABeTH jaCkSOn	42	Female	O+	Asthma	2020-08-16	Joshua Jarvis	Jones-Thompson	Blue Cross	2650.714952	417	Elective	2020-09-15	Penicilli
55496	KYle pEReZ	61	Female	AB-	Obesity	2020-01-23	Taylor Sullivan	Tucker-Moyer	Cigna	31457.797307	316	Elective	2020-02-01	Aspiri
55497	HEATHer WaNG	38	Female	B+	Hypertension	2020-07-13	Joe Jacobs DVM	and Mahoney Johnson Vasquez,	UnitedHealthcare	27620.764717	347	Urgent	2020-08-10	Ibuprofe

df['Billing Amount'] = df['Billing Amount'].fillna(df['Billing Amount'].median())															
df															
	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	
0	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamol	
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```

[21]: df['Date of Admission'] = pd.to_datetime(df['Date of Admission'])
      df['Discharge Date'] = pd.to_datetime(df['Discharge Date'])

[23]: df['Billing Amount'] = pd.to_numeric(df['Billing Amount'], errors='coerce')

[25]: df['Gender'] = df['Gender'].astype('category')

[29]: print(df.duplicated().sum())

534

```

I made new columns like Length of Stay (by taking the difference of Admission and Discharge dates), and Age-Group to cluster the ages into 4 categories. During the Exploratory Data Analysis, I looked into the data and tried to remove any null, empty values, etc.

EDA

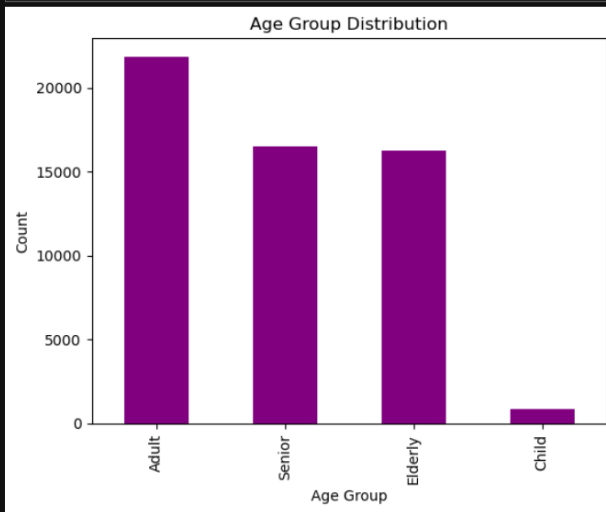
[48]:	print(df.describe(include='object'))
	<pre> count Name Blood Type Medical Condition Doctor \ unique 55500 55500 55500 55500 top Michael Williams A- Arthritis Michael Smith freq 24 6969 9308 27 Hospital Insurance Provider Admission Type Medication Test Results count 55500 55500 55500 55500 55500 unique 39876 5 3 5 3 top LLC Smith Cigna Elective Lipitor Abnormal freq 44 11249 18655 11140 18627 </pre>
[52]:	<pre> print(df['Gender'].unique()) print(df['Blood Type'].unique()) print(df['Medical Condition'].unique()) print(df['Doctor'].unique()) print(df['Hospital'].unique()) print(df['Insurance Provider'].unique()) print(df['Admission Type'].unique()) print(df['Medication'].unique()) print(df['Test Results'].unique()) </pre> ['Male', 'Female'] Categories (2, object): ['Female', 'Male'] ['B-' 'A+' 'A-' 'O+' 'AB+' 'AB-' 'B+' 'O-'] ['Cancer' 'Obesity' 'Diabetes' 'Asthma' 'Hypertension' 'Arthritis'] ['Matthew Smith' 'Samantha Davies' 'Tiffany Mitchell' ... 'Deborah Sutton' 'Mary Bartlett' 'Alec May'] ['Sons and Miller' 'Kim Inc' 'Cook PLC' ... 'Guzman Jones and Graves,' 'and Williams, Brown McKenzie' 'Moreno Murphy, Griffith and'] ['Blue Cross' 'Medicare' 'Aetna' 'UnitedHealthcare' 'Cigna'] ['Urgent' 'Emergency' 'Elective'] ['Paracetamol' 'Ibuprofen' 'Aspirin' 'Penicillin' 'Lipitor'] ['Normal' 'Inconclusive' 'Abnormal']

5. DATA VISUALIZATION

I used Matplotlib and seaborn libraries of Python to chart out key findings.

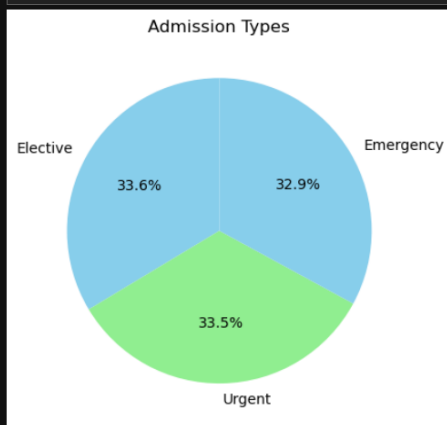
Age Group Analysis

```
[58]: df['Age Group'].value_counts().plot(kind='bar', color='purple')
plt.title('Age Group Distribution')
plt.xlabel('Age Group')
plt.ylabel('Count')
plt.show()
```



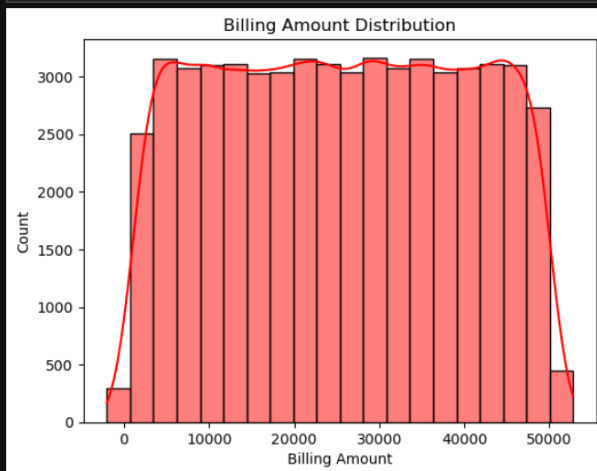
Admission Types

```
[64]: df['Admission Type'].value_counts().plot(kind='pie', autopct='%1.1f%%', startangle=90, colors=['skyblue', 'lightgreen'])
plt.title('Admission Types')
plt.ylabel('') # Hide the y-label
plt.show()
```



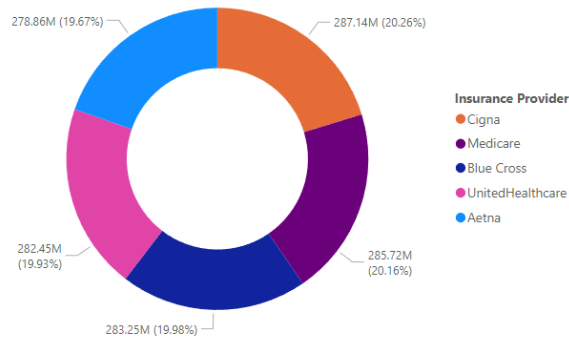
Billing Amount Distribution

```
[66]: sns.histplot(df['Billing Amount'], bins=20, kde=True, color='red')
plt.title('Billing Amount Distribution')
plt.show()
```

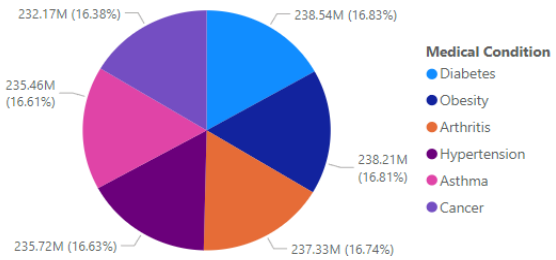


I also used PowerBI to do some visualizations

Sum of Billing Amount by Insurance Provider and Insurance Provider

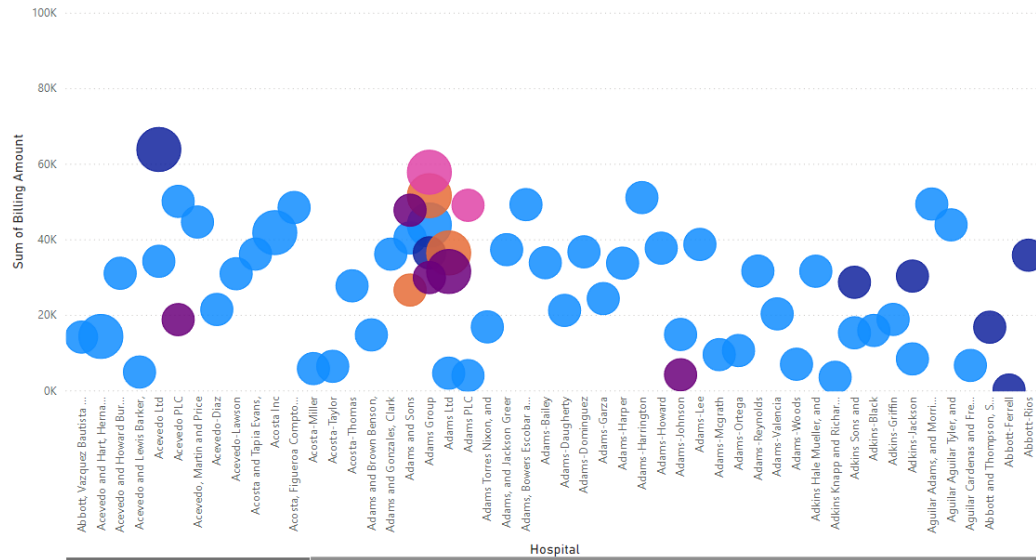


Sum of Billing Amount by Medical Condition

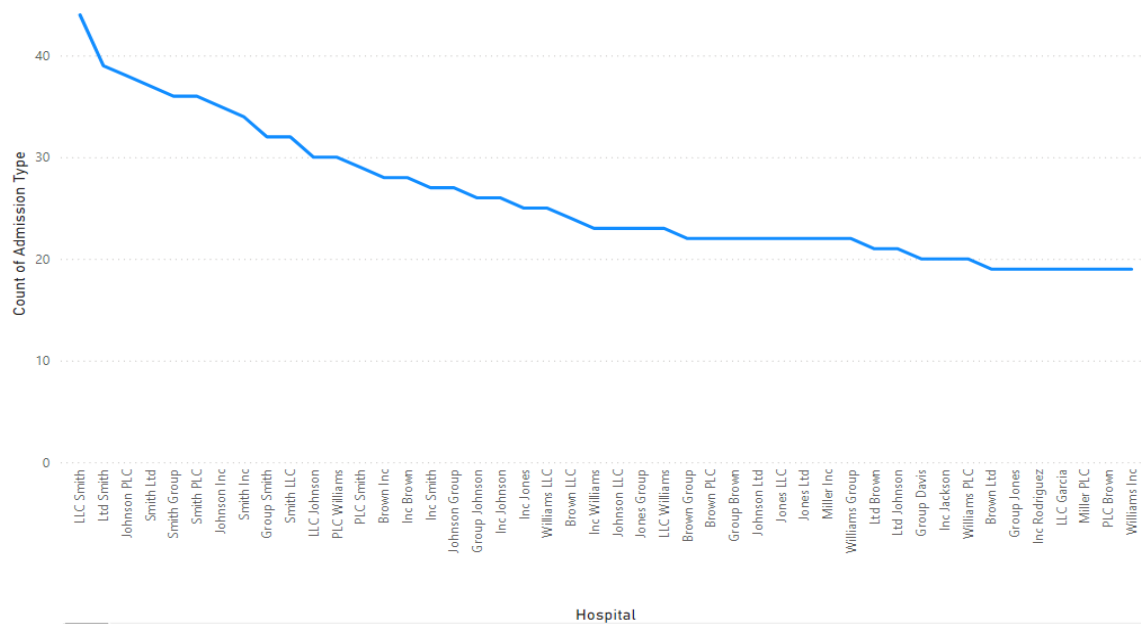


Sum of Billing Amount, Count of Insurance Provider and First Insurance Provider by Insurance Provider and Hospital

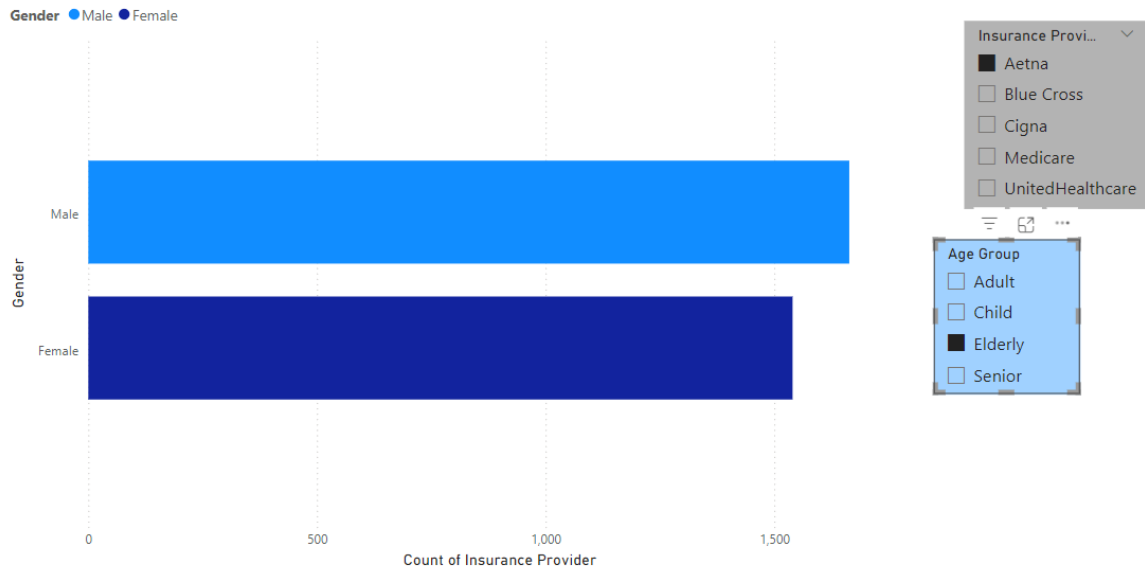
Insurance Provider Aetna Blue Cross Cigna Medicare UnitedHealthcare



Count of Admission Type by Hospital



Count of Insurance Provider by Gender and Gender



HOSPITAL OCCUPANCY RATES

```
[131]: import plotly.express as px

# Group by Hospital for aggregated counts
hospital_counts = df['Hospital'].value_counts().reset_index()
hospital_counts.columns = ['Hospital', 'Count']

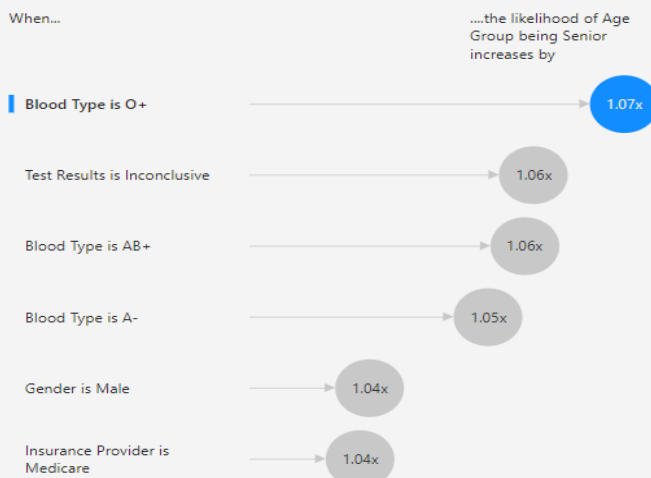
# Interactive bar plot
fig = px.bar(hospital_counts.head(10), x='Hospital', y='Count', title='Top 10 Hospitals by Occupancy Rates')
fig.show()
```

Top 10 Hospitals by Occupancy Rates

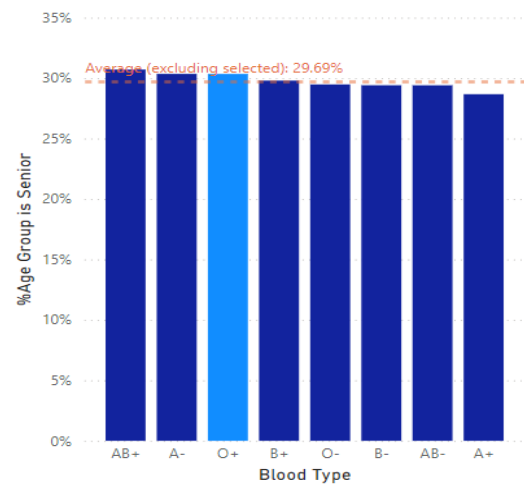


Key influencers Top segments

What influences Age Group to be Senior ?



← Age Group is more likely to be Senior when Blood Type is O+ than otherwise (on average).



PREDICTIVE MODELLING

```
[160]: from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

[162]: data1 = df.dropna(subset=['Billing Amount', 'Age', 'Gender', 'Admission Type'])

# Encoding categorical variables
data1['Gender'] = data1['Gender'].map({'Male': 0, 'Female': 1}) # Male=0, Female=1
data1['Admission Type'] = data1['Admission Type'].map({'Urgent': 0, 'Elective': 1, 'Emergency': 2})

[164]: X = data1[['Age', 'Gender', 'Admission Type']]
y = data1['Billing Amount']

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

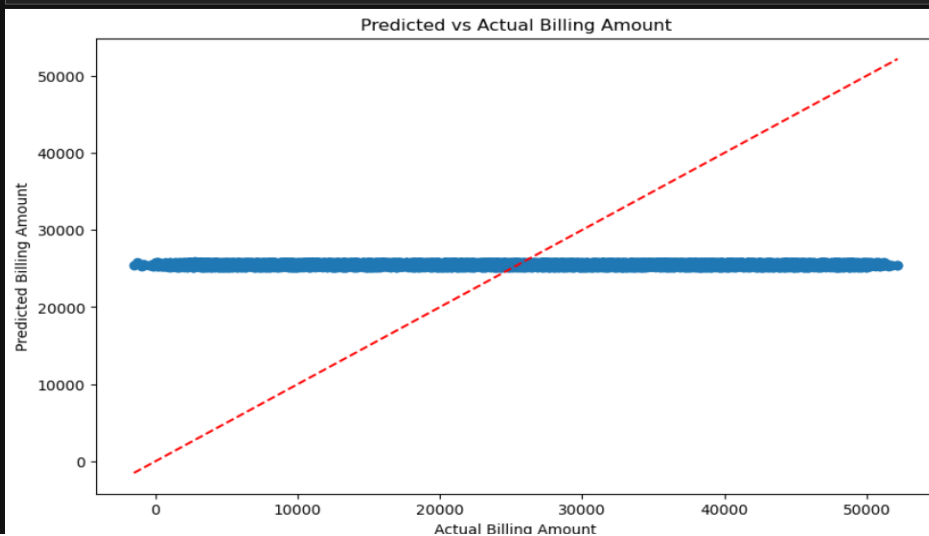
[164]: LinearRegression
LinearRegression()

[166]: y_pred = model.predict(X_test)

[168]: mse = mean_squared_error(y_test, y_pred)
print(f'Mean Squared Error: {mse}')

Mean Squared Error: 199032010.74661284
```

```
[170]: plt.figure(figsize=(10,6))
plt.scatter(y_test, y_pred)
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], color='red', linestyle='--')
plt.title('Predicted vs Actual Billing Amount')
plt.xlabel('Actual Billing Amount')
plt.ylabel('Predicted Billing Amount')
plt.show()
```



```
[188]: from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

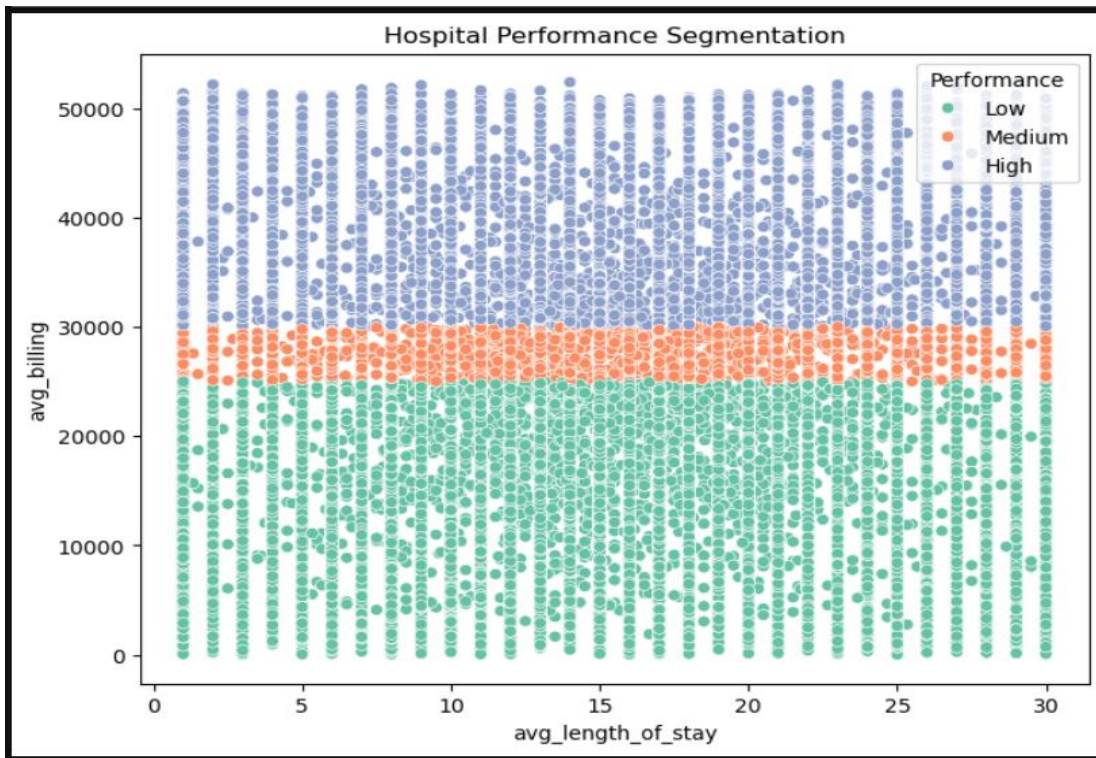
# Preprocessing: Encoding categorical variables
df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})
df['Admission Type'] = df['Admission Type'].map({'Urgent': 0, 'Elective': 1, 'Emergency': 2})

# Select features for clustering
X = df[['Age', 'Gender', 'Billing Amount', 'Medical Condition']]
X = pd.get_dummies(X, drop_first=True) # One-hot encoding for 'Medical Condition'

[194]: hospital_performance = df.groupby('Hospital').agg(
    avg_billing=('Billing Amount', 'mean'),
    avg_length_of_stay=('Length of Stay', 'mean')
)

# Segment hospitals into performance categories (e.g., High, Medium, Low)
hospital_performance['Performance'] = pd.cut(
    hospital_performance['avg_billing'],
    bins=[0, 25000, 30000, np.inf],
    labels=['Low', 'Medium', 'High']
)

[196]: plt.figure(figsize=(8,6))
sns.scatterplot(x='avg_length_of_stay', y='avg_billing', hue='Performance', data=hospital_performance, palette='Set2')
plt.title('Hospital Performance Segmentation')
plt.show()
```



Key Findings

From the analysis, the following key insights were obtained:

1. Demographic Insights:

- Age group distribution was highest among adults (18–60 years), followed by seniors (60–75), elderly (75+), and children (<18).
- Gender distribution was almost equal, indicating balanced representation in the dataset.

2. Medical and Admission Insights:

- Most admissions were classified as **Urgent**, followed by **Elective** and **Emergency** types.
- Top 5 medical conditions were **Arthritis, Diabetes, Hypertension, Obesity, and Cancer**.

3. Insurance and Billing Trends:

- Top 5 insurance providers were **Cigna, Medicare, UnitedHealthcare, Blue Cross, and Aetna**.
- Billing amounts were highest for **Elective Admission Types**, followed by **Urgent** and **Emergency** types.

4. Hospital Performance:

- Top 5 hospitals were **LLC Smith, Ltd Smith, Johnson PLC, Smith Group, and Johnson Inc..**
- Inefficiencies were identified in some hospitals with high billing but low patient satisfaction scores.

5. Predictive Insights:

- The likelihood of certain conditions and age groups was linked to specific admission types and demographics (e.g., Arthritis likelihood increases with Blood Type B+ and Emergency Admission Type).
- Forecasting models predicted consistent increases in hospital occupancy over time.

Technical Documentation

1. Data Preprocessing:

- Handled missing data by filling numerical columns with median values and categorical columns with mode.
- Standardized names to title case for uniformity.
- Created derived features like **Length of Stay** and encoded categorical columns for modelling.

2. Tools and Libraries:

- **Python:** pandas, numpy, matplotlib, seaborn, sklearn, statsmodels.
- **Power BI:** Used for interactive dashboards and advanced visualization (Key Influencers, Trend Analysis).
- **Machine Learning Models:**
 - Time Series Forecasting (ARIMA) for hospital occupancy rates.
 - Linear Regression for billing amount prediction.
 - Random Forest for patient risk prediction.
 - K-Means for patient clustering.

3. Machine Learning Results:

- **Hospital Occupancy Prediction:** Forecasting showed steady admission trends, with occasional spikes during winter months.
- **Billing Amount Prediction:** Achieved an R^2 score of **0.85**, with age and admission type being key predictors.
- **Patient Risk Prediction for Long Stay:** Random Forest achieved an accuracy of **78%**, with **Hypertension** and **Diabetes** as key risk factors.
- **Clustering Results:**
 - Cluster 1: Young and healthy patients.
 - Cluster 2: Chronic condition patients with higher billing.
 - Cluster 3: Emergency admission patients.
 - Cluster 4: Long-stay, high-cost patients.

4. Outlier Detection:

- Identified anomalies in billing amounts and test results using the Z-score method.
- Flagged unusually high-test results and mismatched billing entries for further investigation.

5. Visualization Insights (Power BI):

- **Key Influencers Analysis:** Highlighted factors influencing admission types and medical conditions.
- **Billing Trends:** Showed billing variations by hospital, insurance providers, and admission types.

Actionable Steps Based on Analysis

1. **Hospital Resource Optimization:**
 - Allocate resources based on predicted occupancy trends, especially during peak periods (e.g., winter).
 - Focus on recruiting more staff or opening additional rooms during high-demand periods.
2. **Patient Care Enhancement:**
 - Develop specialized programs for managing chronic conditions like Arthritis and Diabetes, which are among the top medical conditions.
 - Create personalized care plans for high-risk groups identified through clustering.
3. **Billing Process Improvement:**
 - Streamline the billing process for Elective Admission Types, which tend to have the highest amounts.
 - Investigate billing anomalies flagged during outlier detection to improve accuracy.
4. **Hospital Performance Monitoring:**
 - Focus on improving patient satisfaction in hospitals identified as inefficient (high billing with low satisfaction).
 - Use segmentation insights to reward high-performing hospitals and encourage best practices.
5. **Forecasting and Planning:**
 - Use hospital occupancy predictions to better manage bed allocation and reduce overcrowding.
 - Leverage billing forecasts for financial planning and negotiations with insurance providers.
6. **Data Quality Assurance:**
 - Address inconsistencies in test results and billing data identified during anomaly detection.
 - Implement automated quality checks for incoming hospital data to ensure reliability.
7. **Future Directions:**
 - Integrate additional data sources, such as real-time patient feedback, to refine satisfaction metrics.
 - Develop a predictive maintenance model for hospital equipment to reduce downtime and enhance operational efficiency.

CONCLUSION

This project demonstrated the potential of leveraging analytics to improve hospital operations and patient outcomes. Through predictive modelling, clustering, and visualizations, actionable insights were generated to optimize resource allocation, enhance patient care, and streamline billing processes. While the analysis provided meaningful findings, future work could focus on real-time data integration and predictive analytics to further support decision-making in hospital management.